7PAM2002 Data Science Project Module

Choosing a Project Form

Semester A 2025/2026

This form will be used by your supervisor to agree your project topic and dataset. Complete as much of the form as possible then submit the form into 'Assignments' on the Project Module Canvas site.

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Course: 2 year Advanced Research
Semester intake to the course: B

SECTION A	
What is your proposed project title or topic?	**Machine learning–based intelligent traffic congestion prediction and optimisation using real-time urban mobility and sensor data**
What is the Research Question for your project?	1. Which spatiotemporal features most effectively support accurate traffic congestion prediction using real-time machine learning models? 2. How accurately can machine learning models predict traffic congestion under dynamic and heterogeneous urban traffic conditions? 3. How can machine learning-based predictions support traffic optimisation strategies within intelligent transportation systems?
Dataset website address (or organisation and person the dataset is from).	**Dataset 1:** https://zenodo.org/records/3939793
Where was the data originally collected? (who, when ,where)	Yixian Chen Xinyu Chen Published July 11, 2020 A Large Scale PeMS Traffic Speed Dataset traffic speed data set collected by California Freeway Performance Measurement System (PeMS). In this data set, it has 11160 time series corresponding to 11160 sensors/locations.

What type of data are you using? (e.g. image/ tabular/category/continuous etc)	Tabular.
SECTION B	
Can you attend all 6 supervision sessions on-campus? If not state reason.	YES, I'll definitely attend to all the sessions.
What was the subject of your BSc degreeand any other Master's degrees you have taken?	Bachelor's degree in Computer Science.
What are your career aims and/or the industry sector you would like to get into?	I aim to pursue a career as a Data Scientist or Machine Learning Engineer, with a strong interest in predictive analytics.
A brief account of your programming/data science experience (including work or placement).	I have a solid understanding of the Python data stack; for more complicated modeling tasks, I use TensorFlow, and my go-to tools for data manipulation are pandas and NumPy. I have a solid foundation in statistics thanks to my academic background, so I enjoy exploring the "why" behind the data through exploratory analysis rather than just creating models.
Any work experience (including non-computing related).	No.
Do you have any hobbies, activities or interests.	I am interested in data analysis, machine learning research, reading about emerging AI technologies, and continuous skill development through online courses and projects.